# How Many Objects Are Created by JVM

## (in a Normal Environment, Including Thread Objects)

**Example:**

```
public class Test {
    public static void main(String[] args) {
}
```

### MINIMUM OBJECTS CREATED BY JVM (Normal (Single-Threaded) Java Environment)

| # | Object Created By JVM | How Many | Why It Is Created |
|---|---|---|---|
| 1 | **Class object** (Class<Test>) | 1 | Represents loaded class |
| 2 | String object for class name | 1 | Stored in String Pool |
| 3 | Main Thread object | 1 | To execute: 'main() |
| 4 | **ThreadGroup** object | 1 | Loads class |
| 5 | **ClassLoader** object | 1 | Loan class |
| 6 | System Class objects | Multiple | 'Object, 'String,*, etc.) |
| 7 | Security / Runtime objects | Multiple | JVM runtine support |

**Minimum Guaranteed User-Visible Objects = 5**

### JVM vs Developer Created Objects

| Category | Normal Environment |
|---|---|
| JVM Internal Objects | ✔ Many |
| **User-Created** Objects ("**new**') | ✘ **No** |
| **Thread Objects** | ✔ **1** |
| **Class Objects** | ✔ **1** per class |
| String Literals | ✔ Depends on Code |

### JVM-Created Objects Categorized

**Execution**
Thread, ThreadGroup

**Memory**
String Literals

**Security**
SecurityManager (Optional)

**Loading**
Class, ClassLoader

**Core Runtime**
Object, String, System

**Security**
SecurityManager

- ✘ Additional Thread Objects ⊗
- ✘ Runnable Objects ⊗
- ✘ User Objects ⊗
- ✘ Executor Services ⊗

**MINIMUM JVM-CREATED OBJECTS** IN A Normal Environment = **5 +**

Core Java Classes, **Main Thread, ClassLoader, ThreadGroup, and Hidden Runtime** Support Are **Always Created By JVM.**

Excellent question — this goes **deep into JVM internals**, and interviewers love asking it in tricky ways 
Let's answer it **precisely and cleanly in table form**, including **thread objects**, exactly as you asked.

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**1 Minimum Objects Created by JVM**

**In a Normal (Single-Threaded) Java Environment**

Example:

```
public class Test {
   public static void main(String[] args) {
   }
}
```

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**2 JVM-Created Objects (Always Created)**

| # | Object Created by JVM | How Many | Why It Is Created |
|---|---|---|---|
| **1** | **Class object (Class<Test>)** | 1 | Represents loaded class |
| **2** | **String object for class name** | 1 | Stored in String Pool |
| **3** | **Main Thread object** | 1 | To execute main() |
| **4** | **ThreadGroup object** | 1 | Manages main thread |
| **5** | **ClassLoader object** | 1 | Loads class |
| **6** | **System Class objects** | Multiple | Object, String, etc. |
| **7** | **Security / Runtime objects** | Multiple | JVM runtime support |

 **Minimum guaranteed user-visible objects = 5**

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**3 JVM Objects vs Developer Objects**

| Category | Normal Environment |
|---|---|
| **JVM internal objects** | ✅ Many |

| | |
|---|---|
| **User-created objects (new)** | ✗ Zero (if no new) |
| **Thread objects** | ✓ 1 (main thread) |
| **Class objects** | ✓ 1 per class |
| **String literals** | Depends on code |

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### 4 Thread-Related Objects (Always Present)

| Object | Count | Notes |
|---|---|---|
| **Thread** | 1 | Main thread |
| **ThreadGroup** | 1 | Main group |
| **Stack frame objects** | Multiple | Per method call |
| **Runnable** | 0 | Not needed for main |

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### 5 JVM-Created Objects – Categorized View

| Category | Objects Created |
|---|---|
| **Execution** | Thread, ThreadGroup |
| **Loading** | Class, ClassLoader |
| **Memory** | String literals |
| **Core Runtime** | Object, String, System |
| **Security** | SecurityManager (optional) |

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### 6 What Is NOT Created in Normal Environment

| Thing | Created? |
|---|---|
| **Additional Thread objects** | ✗ No |
| **Runnable objects** | ✗ No |

| | |
|---|---|
| **User objects** | ❌ No |
| **Executor services** | ❌ No |

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**7 Interview-Ready Final Table (One Look Answer)**

| Question | Answer |
|---|---|
| **Minimum JVM-created objects** | **5+** |
| **Thread objects** | **1 (main)** |
| **Class objects** | **1 per loaded class** |
| **User objects** | **0 (if no new)** |
| **JVM internal objects** | **Many (hidden)** |

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**□ One-Line Interview Answer**

**Even in a normal single-threaded Java program, the JVM creates multiple objects—including at least one Class object, one main Thread object, a ThreadGroup, and runtime support objects—before user code executes.**

If you want next:
✅ JVM startup object creation flow (step-by-step)
✅ Diagram: JVM → ClassLoader → Thread → main()
✅ Exactly which objects are created first
✅ Trick interview questions with correct answers